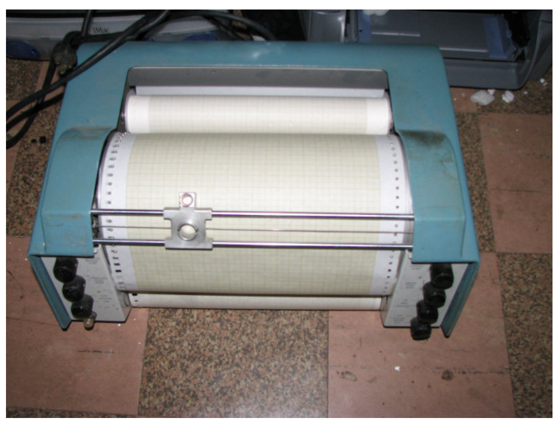
Build and Installation Instructions

1627 Plotter attachment to IBM 1130

Plotter models this board can be used with

This board is designed to be used with the IBM 1627 model 1 or IBM 1627 model 2 plotters. These are private labeled versions of the Calcomp 565 and Calcomp 563 respectively.



Selecting the plotter model

To support the larger plotters - 1627 model 2 or Calcomp 563 - install a jumper across the two pins marked ‘model 2’ on the PCB. The jumper slows down the response to pen movements from 3.8 milliseconds to 5.8 milliseconds due to the larger drum size of those models. If the plotter is an IBM 1627 model 1 or a Calcomp 565, no jumper is installed on the pins.

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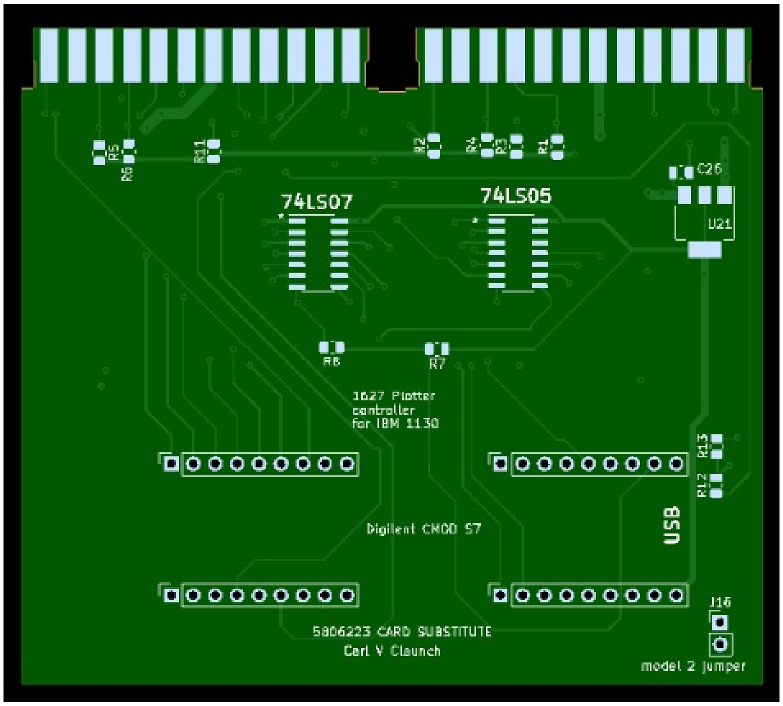
Building the card



It is recommended that the contacts be installed onto the printed circuit board before adding the other components. See the following section for how to source the contacts and how to solder them onto the board.

Next solder the resistors, capacitors and three integrated circuits on the PCB. The last step will be to install the daughterboard onto the PCB. Push the Digilent CMOD S7 FPGA board through the holes, ensuring that you have the USB connector end oriented properly. This board must lay low on the PCB to avoid mechanical interference with other SLT cards when it is in the compartment. Solder the FPGA board in place and trim the leads close to the back of the main PCB.

If not done already, the plastic covers are inserted over the two sockets and snapped into place. The card would then be ready for bench testing, installation in the IBM 1130 and testing on the system.

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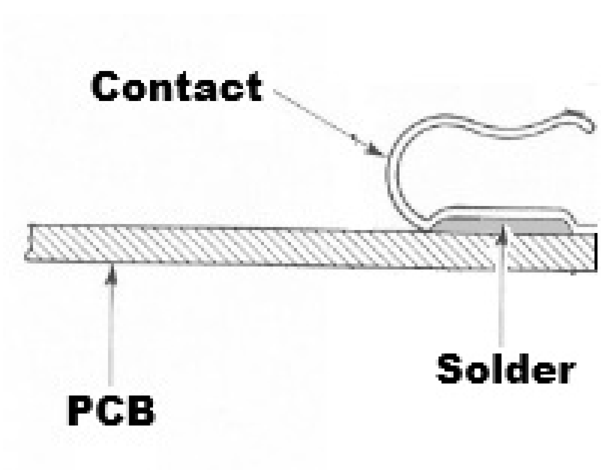
<u>Installing contacts and cover on the substitute card</u>

The card has space for 48 contacts to form the two 2x12 slot plugs. Generally the best way to get the contacts is to disassemble a donor Solid Logic Technology (SLT) card.

Remove the plastic cover over the 2x12 pin socket from which you can extract 24 contacts. The edges of the card slope outward toward the end which tends to hold the plastic cover in place. A small screwdriver can be used to bend the sides of the covers to allow them to be pulled off the card. Also save the plastic cover for re-use on the new card.

Unsolder contacts from the donor card(s) until you have the 48 needed for the plotter substitute controller card. A hot air rework tool on a heated rework table works best for this, but use any method you have that works.

The contact has a flat plate that solders onto the PCB pad and continues bending around and upward to provide a springy bit of metal that the SLT backplane pin can push against. Do not install the contact with the bend at the edge of the card, the open U shape should be at the edge.



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First, using a solder paste with a higher melting point (e.g. 183 C), solder down the contacts on one side. Set the rework table to about 150 C and a hot air rework gun to over 183 C.

Lay solder paste on the pads on the side of the card you are working with first, then place all the contacts on that side sitting in roughly the correct position atop the paste.

Using the hot air rework gun, melt the pads while verifying that the contact slides into proper alignment once the solder has fully flowed. Work from one side to the other, until you have all the contacts lined up correctly and soldered to the board. A clamp or clip might be placed on the contact portion directly on the PCB pad, with the other side of the clamp/clip on the underneath portion of the PCB. That would assure better alignment but the key will be to heat long enough to be certain the solder paste is fully melted underneath the contact.

Using a second lower heat solder paste (e.g. 138 C), turn the rework table down to about 115C and the hot air rework gun to just a bit over 138 C. Once the table is at the proper temperature, lay the card with the other side's pads facing you. Lay solder paste on the pads and place the remaining contacts in approximately the proper spot. If using a clamp or clips, add them now.

Using the hot air rework gun, melt the pads while verifying that the contact sits in proper alignment once the solder has fully flowed. Work from one side to the other, until you have all the contacts lined up correctly and soldered to the board.

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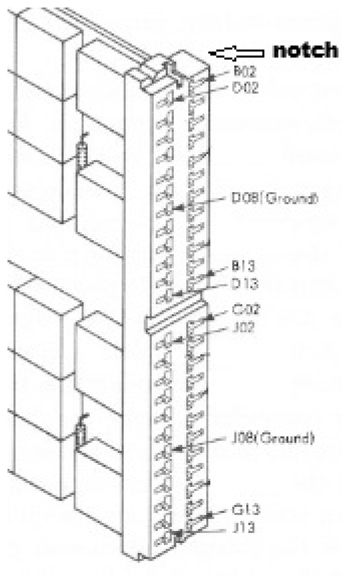
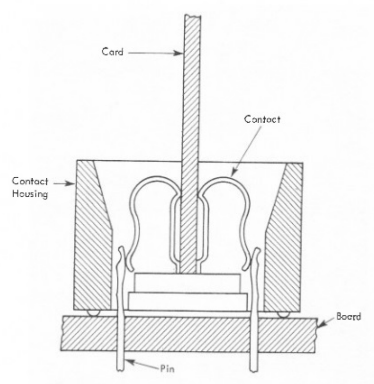
Slide the plastic covers over the contacts to finish the SLT slot plugs. The notch on the cover is on the side of the PCB away from the chips.



Do a test insertion into a vertical pair of open SLT slots in the 1130 to confirm that the card does snap down into place.



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Load the logic into the FPGA

Load the CMOS S7 FPGA board with the memory configuration file to activate the logic for the plotter controller functionality. I use Xilinx Vivado, doing an autoconnect to the board through a USB cable, selecting the flash RAM that is listed as part of the discovered board, then programming the FPGA with the memory configuration file.

If the flash RAM chip on the CMOD S7 board is a MX25L3233F, then the FPGAload.mcs file included in the github project can be downloaded, otherwise a project must be built in Vivado to synthesize and generate the bitstream from the source files, then creating a memory configuration file to download to the RAM chip type that is built into your CMOD S7 board. A file on github explains how to generate IP and add additional source files to build the memory configuration file.

If you might ever use the USB serial link to capture movement commands from the card, there is a hardware modification needed to the CMOD S7 board (see later in this document) to safely connect when the card is in the IBM 1130.

Installing the substitute 5806223 controller card in the 1130

Once you are done with any visual inspection, continuity testing and possible bench testing, slide the double width card into slots M4 and M5 of Gate A, Compartment C1. Push down until it is securely in place.

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<u>Three alternative approaches to connecting the 1627 cable</u>

A - The IBM 1130 system already has cabling installed between Gate A, Compartment C1, Slot N5 and slot PF6 of the SMS sockets for peripherals inside the 1130 system

B - you will wire up a cable between SMS socket slot PF6 and Gate A, Compartment C1 Slot N5

C - you will wire up a modern connector near Gate A, Compartment C1 into which the plotter signal cable will plug

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Alternative A - using existing wiring in an 1130 system

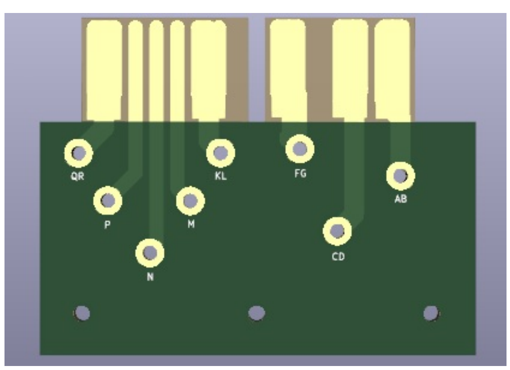
Since the 1130 already has wiring from a socket in slot PF6 of the SMS peripheral connection block over to Slot N5 of Gate A, Compartment C1, you only need to connect the wiring from the plotter connector to an SMS paddle card that will be plugged into slot PF6.

A female 19 pin connector, Cannon SK19-23C, plugs into the rear of the plotter. It carries all the power and signal lines between the plotter and the 1130 system.

Create an SMS power paddle card and run 18 gauge stranded wire to it from the 19 pin connector:

Cannon pin 15 is connected to pad P of the power paddle card
Cannon pin 17 is wired to pads F and G of the power paddle card
Cannon pin 18 is wired to pads C and D of the power paddle card
Cannon pin 14 is wired to pads K and L of the power paddle card

A SMS power paddle card has a notch cut in the card approximately where pads H and J would be on a normal SMS signal paddle card. This ensures that signal cards CANNOT fit into the power SMS sockets on the 1130 system. Gerber/drill files are provided to make this card as well as files to make an SMS signal paddle card used below.



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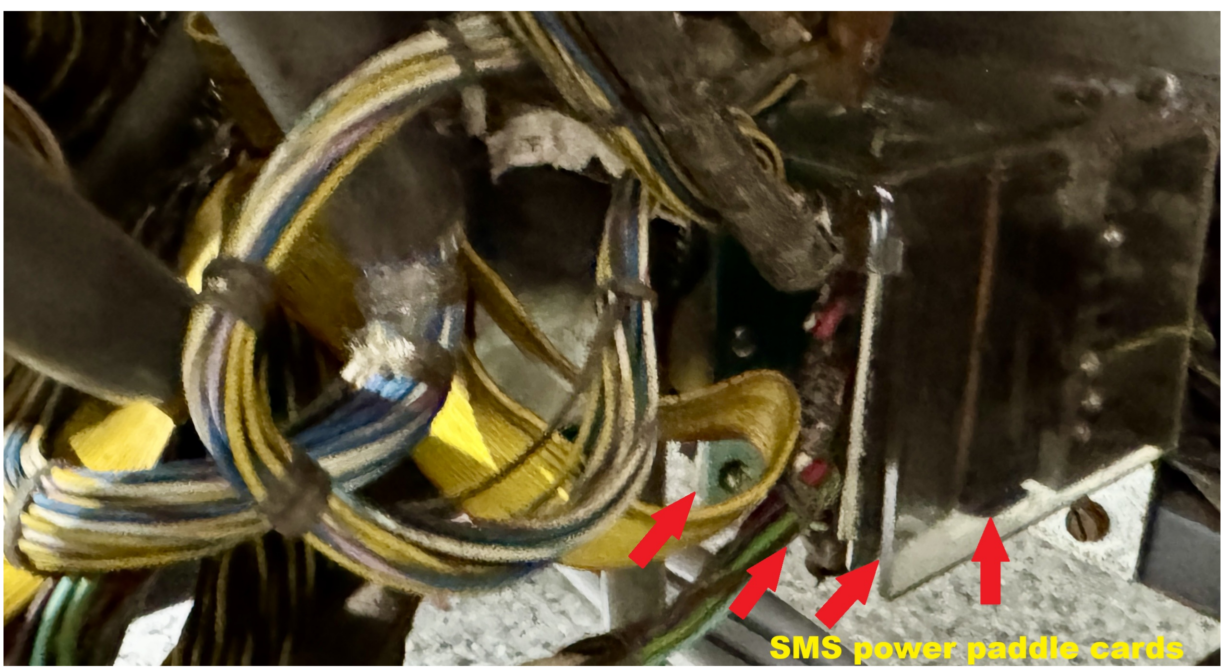
1627 Plotter attachment to IBM 1130

A regular SMS paddle card, used for the signal connections, is connected as follows using smaller gauge stranded wire:

> Cannon pin 5 is wired to the signal paddle card PF6 pad R
> Cannon pin 6 is wired to the signal paddle card PF6 pad Q
> Cannon pin 7 is wired to the signal paddle card PF6 pad N
> Cannon pin 8 is wired to the signal paddle card PF6 pad P
> Cannon pin 11 is wired to signal paddle card PF6 pad M
> Cannon pin 12 is wired to signal paddle card PF6 pad L
> Cannon pin 16 is wired to signal paddle card PF6 pad G
>
> Install a 3.9K resistor connected between PF6 pad G and PF6 pad C

Open the lower SMS socket block inside the IBM 1130, where peripheral power connections are made. Insert the power paddle card you manufactured into slot PP2 just after the paddle card that powers the 1053 console printer. Close up the power socket block.



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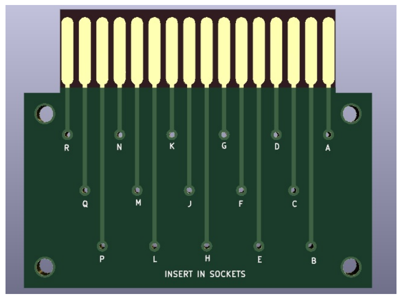
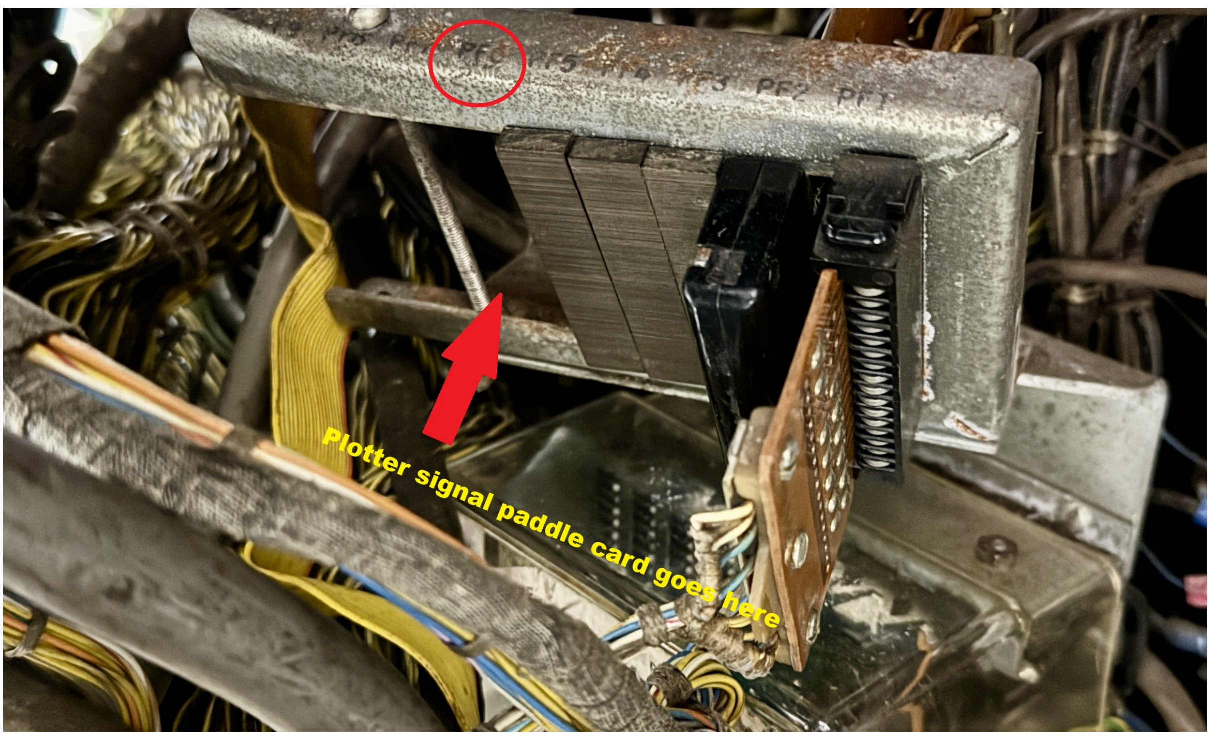
Open the upper SMS socket block that connects signal lines to peripherals. Insert the signal paddle card into slot PF6 of this block. Close up the socket block.

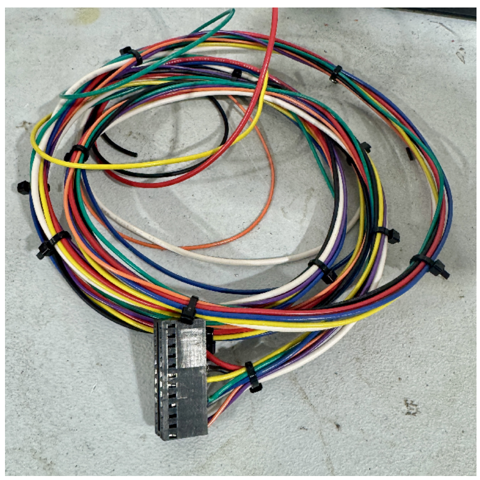




Alternative B - Adding wiring that matches the IBM approach

Construct the two paddle cards and the cable to the Cannon 19 pin connector using the instructions from Alternative A above. The additional work that must be done is to add a wire bundle between the SMS signal socket block slot PF6 and Gate A, Compartment C1, Slot N5 where the signals are connected to the Solid Logic Technology (SLT) backplane for that compartment.

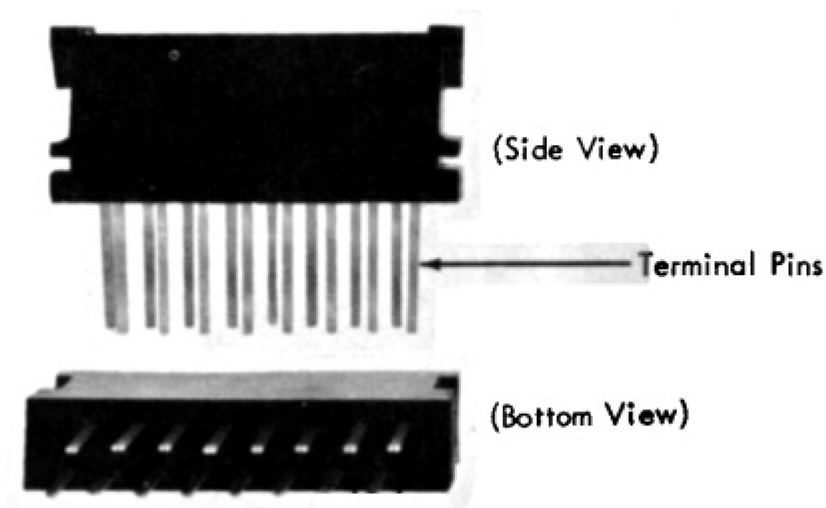
Build an SLT connector that will plug into slot N5 - a source is provided in the bill of materials.



Using appropriate twisted pairs, routing one wire of each pair to the 1130 signal ground, connect:

Paddle socket position R is wired to N5 connector position B02
Paddle socket position Q is wired to N5 connector position D02
Paddle socket position N is wired to N5 connector position D04
Paddle socket position P is wired to N5 connector position B03
Paddle socket position M is wired to N5 connector position B04
Paddle socket position L is wired to N5 connector position D05
Paddle socket position C is wired to N5 connector position B10

Install the female SMS socket in space PF6 of the upper SMS socket block. Plug the SMS signal paddle card you wired from the plotter (as covered in alternative A) into this socket. Close up the SMS signal socket block.



Plug the SLT connector end you just wired to the SMS female socket into Gate A, Compartment C1, Slot N5. See the instructions further down for how to do this.

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<u>Alternative C - modern connector for the signal connections</u>

Use the instructions from alternative A to build an SMS power paddle card and wire it to the Cannon 19 pin connector. Install the SMS power paddle card in the lower SMS socket block as described in alternative A.

Buy any 8 pin male and female socket pair. Install the female socket on the 1130 gate A near Compartment C1. Build the SLT socket as described in alternative B.

Wire the 19 pin connector to the male socket:

- Cannon pin 5 is wired to the socket pin 1
- Cannon pin 6 is wired to the socket pin 2
- Cannon pin 7 is wired to the socket pin 3
- Cannon pin 8 is wired to the socket pin 4
- Cannon pin 11 is wired to the socket pin 5
- Cannon pin 12 is wired to the socket pin 6
- Cannon pin 16 is wired to a 3.9K resistor whose other end is wired to the socket pin 7
- Cannon pin 15 is wired to the socket pin 8 (also wired to the power SMS paddle card)

Connect the female socket:

- socket pin 1 to N5 connector position B02
- socket pin 2 to N5 connector position D02
- socket pin 3 to N5 connector position D04
- socket pin 4 to N5 connector position B03
- socket pin 5 to N5 connector position B04
- socket pin 6 to N5 connector position D05
- socket pin 7 to N5 connector position B10
- socket pin 8 to signal ground below the card compartment

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Installing the SLT connector in slot N5

Remove the card cover from compartment C1 of gate A. Loosen the two screws circled in red below that hold the right side support for the cover.

Plug the SLT connector into slot N5 and pass the wires to the right so that they will be held under the right side support when you re-install that part.

Secure the cable from the male connector where it exits the 1130 system to protect from someone accidentally pulling the connectors. I used twist ties to anchor the cable inside compartment C1 after the connector was plugged in, as well as anchoring the cable as it runs through the 1130 and at the exit point where the cable comes out of the machine.

Cable passes behind the hinge and up between the two gates A and B.

Cable anchored to metal finger just behind the side support.

Building an SLT cable connector to wire in the plotter

This cable will can be built two ways - so that it can be hooked to the SMS signal connector block inside the 1130 allowing the SMS paddle card to be used to hook up the plotter, or directly to the connector that fits on the plotter. For alternatives B or C, you must create a connector to plug into Slot N5 of Gate A, Compartment C1 of the 1130. Alternative A is for a system with the connector already in place and wired, thus skip this step.

Using the parts listed in the Bill of Materials, crimp the socket pins onto the wire ends and insert the pins into the plastic housing in the appropriate spots.

Testing the card with a bench setup

The file Bench testing setup.txt covers the process of building a test set-up on the bench and testing the card before inserting it into the IBM 1130. This step can be skipped if you are comfortable that the card is properly constructed and free of short circuits.

Testing the plotter with the 1130

With the plotter connected but power off, execute an XIO instruction pointing to an IOCC that specifies a Sense Device operation and area code 5. The result in the accumulator register (ACC) will be the Device Status Word (DSW). Bit 15 should be 1, indicating that the controller card is present but the plotter is not ready to accept commands.

Turn on the plotter power and issue the same XIO instruction. The DSW should have all bits off, indicating that the plotter is ready to be used.

Set up an XIO instruction pointing to an IOCC that specifies a Write operation, area code 5, and points to a word which has the intended movement command. The six leftmost bits of that word specify the movement you wish the plotter to make, with a 1 in the position indicating that movement should be made.

Bit 0 requests that the pen be lowered onto the paper
Bit 1 requests that the drum turn down one position
Bit 2 requests that the drum turn up one position
Bit 3 requests that the pen move right one position
Bit 4 requests that the pen move left one position
Bit 5 requests that the pen be raised off the paper

In single step mode, execute that XIO instruction with your desired movement. Verify that the plotter moved as expected. Verify that the interupt level 3 light on the console display pedestal is illuminated.

The next address executed will be the interrupt handler for level 3. In memory location 000B is an address for the interrupt handler. The hardware will store the address from which we were interrupted into the first word of the interrupt handler and then execute the instruction in the second word of the handler.

If you press the RESET button on the 1130 console, the interrupt request for level 3 should go away.

Next, code two XIO instructions and a Halt instruction in sequence. Open the top cover of the 1130 and turn on the Interrupt Delay switch. The first XIO points to the IOCC which specifies Write to area 5 pointing at a word with your desired movement set in bits 0 to 5. The second XIO points to an IOCC which specifies Sense Device with area 5.

Set the 1130 into run mode, point the IAR at the first XIO instruction, and push the PROG START button on the console. The plotter should move as you requested, the interrupt request lamp for level 3 should be illuminated and the ACC should display a DSW with bits 0 and 14 turned on. This indicates operation complete (0) and that the plotter was still busy at the time we issued the Sense Device.

Set the IAR to the second XIO instruction and push PROG START again. The ACC should now display a 1 only in bit 0, operation complete, but without the busy bit (14) turned on.

Set up an XIO instruction pointing to an IOCC specifying a Sense Device for area 5. Bit 15 of the second word of the IOCC should be set to 1. This specifies reset bit 15. Set the 1130 mode to Single Instruction and set the IAR to this XIO instruction. Push the PROG START button.

The interrupt request lamp for level 3 should be turned off. The DSW should have all 0 bits indicating that the plotter is now ready for the next command and that the operation complete was reset.

Try all six of the bits for movement commands, verifying that each on works as intended. That should be sufficient to verify that the controller card is working properly. Optionally one could load the IBM diagnostic program for the 1627 plotter and run it as described in the A0x ALD that covers this diagnostic program.

Connecting an optional USB serial link cable

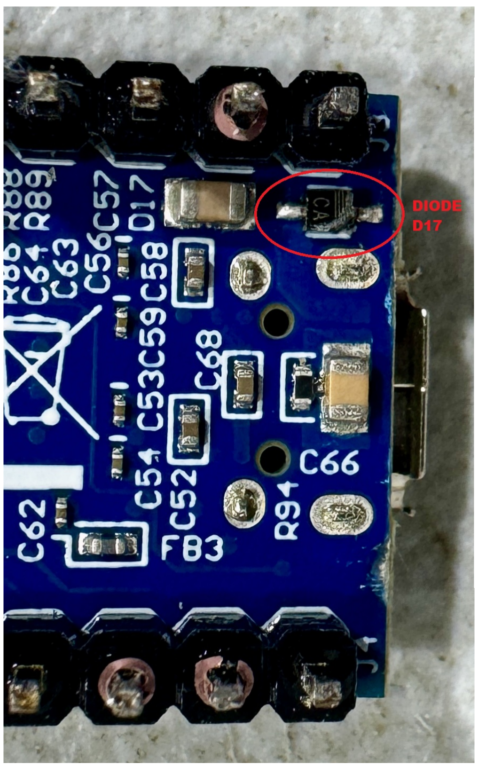
A mini USB connector on the board provides a stream of ASCII lines, one per XIO Write issued to the plotter. Route the cable from the end of the card out of Compartment C1 and out of the IBM 1130 system.

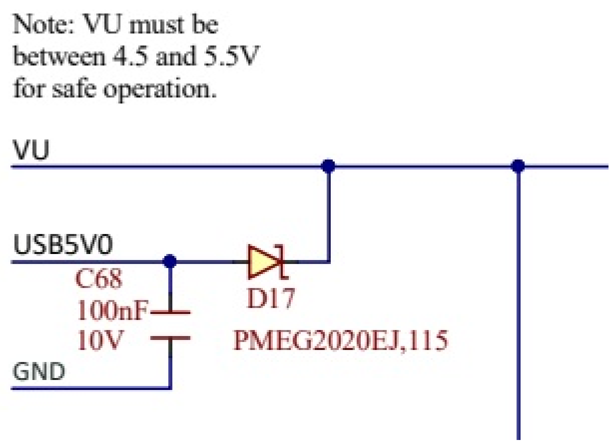
Hooking the other end of the cable into a PC, Mac or workstation that supports serial over USB provide the ability to capture every movement issued to the plotter. Software in that external computer can interpret those movements to draw the same image as is appearing on paper in the plotter.

This may draw the image in real time on a screen or produce a graphic file. The data is in ASCII, running at 38,400 baud, 8 bit, no parity, 1 stop bit. Each line is a set of six ASCII 0 or 1 characters for the six bits sent by the XIO Write command, followed by a CR and then an NL character. For example, a command to lower the pen onto the paper would output this line - **100000 (nl) (cr)** on the serial link.

Modifying the CMOD S7

To eliminate the issue of dual powering for the CMOD S7 board - the regulator on my card and the USB connection - we will remove diode D17 from the bottom of the CMOD S7, which ensures that USB power is never applied to the CMOS S7 nor to the controller card on which it is mounted.





If new logic needs to be downloaded to the configuration memory device on the CMOD S7 board, the controller card must be removed from the IBM 1130. A bench supply should deliver +12V to pin G02 of the card, with ground connected to both pins D08 and J08. The USB cable can then be plugged in and Vivado used to update the new logic into the memory chip.